
Exotic Options Pricing by Using Martingale Optimal Transport

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1 Robust Option Pricing: Financial Formulation of the Problem

Exotic options are derivative contracts whose characteristics differ from vanilla options (standard calls and puts) through more complex payment mechanisms, exercise conditions, or underlying assets. Unlike vanilla options that can be valued using closed-form formulas like Black–Scholes, exotic options often require advanced numerical methods for their evaluation.

1.1 Typology of Exotic Options

Exotic options can be divided into several categories:

- **Path-dependent options:** Asian options (based on averages), lookback options (based on maximum or minimum), barrier options
- **Multi-asset options:** Basket options, rainbow options, quanto options
- **Complex payoff options:** Digital/binary options, power options, compound options

Example

Asian options are evaluated based on the average price of the underlying during a specified period, thus reducing volatility and the influence of short-term market manipulations. Lookback options allow the holder to “look back” and exercise the option at the most favorable price reached during the contract’s lifetime. Barrier options activate or deactivate when the underlying price reaches a predefined level.

The valuation of exotic options presents several challenges:

- Absence of analytical formulas in most cases
- Sensitivity to assumptions about the underlying asset dynamics
- Path dependency requiring complex simulations or numerical schemes
- Parameter uncertainty (volatility, correlations)

Parameter uncertainty is particularly problematic. For example, future volatility is unknown and must be estimated, which introduces significant model risk. Similarly, correlations between assets are unstable and difficult to predict accurately, which complicates the valuation of multi-asset options.

These challenges highlight a key limitation of classical pricing approaches. In practice, these assumptions are difficult to validate and may lead to significant pricing errors. This motivates the use of robust (or model-independent) methods, which aim to derive price bounds using only observable market information and no-arbitrage principles.

1.2 Setup and Standing Assumptions

Throughout this work, we consider the following assumptions:

- **Two maturities:** We consider a discrete-time model with two future dates, $t = 1$ and $t = 2$, and a risky asset with prices S_1 and S_2 .
- **Zero interest rates:** We assume, for simplicity, that the risk-free rate is zero, so that discounting plays no role.
- **Full call surface observable:** For each maturity $t \in \{1, 2\}$, European call options with all strikes $K \geq 0$ are liquidly traded. Their market prices are denoted

$$C_t(K) = \text{price at time 0 of } (S_t - K)^+.$$

- **No arbitrage:** The market is free of arbitrage opportunities.

1.3 From Call Prices to Marginal Distributions

A fundamental observation is that knowing the call prices $C_t(K)$ for all strikes K completely determines the distribution (law) of S_t . Therefore, the market provides full information about the marginal distributions of S_1 and S_2 .

Explanation

This is because call option prices encode the distribution of the underlying asset. Indeed, for a fixed maturity t , the function $K \mapsto C_t(K)$ contains information about how likely it is for S_t to exceed different strike levels.

More precisely, the derivative of the call price with respect to the strike satisfies

$$\frac{\partial C_t(K)}{\partial K} = -\mathbb{P}(S_t > K),$$

so that call prices determine the tail distribution of S_t .

We denote by μ_1 and μ_2 the marginal distributions of S_1 and S_2 respectively, as implied by observed market call prices. The key point is that *the market provides complete information about these marginal distributions, but says nothing about the joint distribution of (S_1, S_2)* . It is precisely this residual uncertainty about the joint law that robust pricing aims to account for.

1.4 The Robust Pricing Framework

Since the market does not fully determine the joint law of (S_1, S_2) , there is no unique risk-neutral measure. Instead, no-arbitrage implies the existence of a pricing measure Q belonging to the set

$$\mathcal{M}(\mu_1, \mu_2) := \left\{ \mathbb{P} \text{ on } \mathbb{R}_+^2 \left| \begin{array}{l} S_t \sim \mu_t \text{ for } t = 1, 2, \\ \mathbb{E}^{\mathbb{P}}[S_2 | S_1] = S_1 \text{ a.s.} \end{array} \right. \right\},$$

under which the discounted price process is a martingale. The martingale constraint reflects the no-arbitrage condition. Fixing the marginals μ_1 and μ_2 encodes all the information the market provides, while the remaining freedom in choosing the joint law captures model uncertainty.

We are interested in a contingent claim with payoff at time 2 of the form

$$C(S_1, S_2),$$

where $C : \mathbb{R}_+^2 \rightarrow \mathbb{R}$ is a given measurable function. Such a payoff depends on the joint evolution of the asset at both dates and is typically not directly traded in the market. Because no unique pricing measure exists, the claim has no unique price; instead, we seek a *price bound* via super-replication.

1.5 The Super-Replication Problem

The objective is to determine the smallest initial capital required to hedge the payoff $C(S_1, S_2)$ in a model-independent (robust) way.

To this end, we consider semi-static trading strategies consisting of:

- a static position in call options with maturity $t = 1$,
- a static position in call options with maturity $t = 2$,
- a dynamic trading strategy in the underlying asset between times 1 and 2.

This structure is rich enough to hedge a wide class of exotic payoffs while remaining consistent with available market instruments.

Such a strategy generates a terminal payoff of the form

$$\phi(S_1) + \psi(S_2) + \Delta(S_1)(S_2 - S_1),$$

where:

- ϕ is a function generated by a portfolio of calls at time 1,
- ψ is a function generated by a portfolio of calls at time 2,
- $\Delta(S_1)$ represents the number of shares held between times 1 and 2, chosen after observing S_1 .

In order to hedge robustly, we require the portfolio to dominate the payoff *for every possible scenario*: $\phi(S_1) + \psi(S_2) + \Delta(S_1)(S_2 - S_1) \geq C(S_1, S_2)$ for all $(S_1, S_2) \in \mathbb{R}_+^2$.

This pointwise inequality represents a worst-case hedging requirement: the strategy must cover the payoff regardless of how the market evolves. The super-replication price is therefore a conservative upper bound on the fair value of the claim.

The initial cost of the strategy is $\int \phi d\mu_1 + \int \psi d\mu_2$ as the dynamic leg Δ contributes no initial cost: it involves self-financing trades in the underlying asset, so gains and losses arise only from price movements between times 1 and 2.

Robust Pricing Problem

The robust super-replication price of the claim is defined as the cheapest admissible strategy:

$$\inf_{\phi, \psi, \Delta} \left\{ \int \phi d\mu_1 + \int \psi d\mu_2 \mid \phi(S_1) + \psi(S_2) + \Delta(S_1)(S_2 - S_1) \geq C(S_1, S_2) \quad \forall (S_1, S_2) \right\}.$$

2 From Super-Replication to Martingale Optimal Transport

We now show that the super-replication problem admits an equivalent probabilistic dual formulation, which leads to a *martingale optimal transport* (MOT) problem. This duality is the cornerstone of the robust pricing approach.

2.1 From Pointwise Inequality to Expectations

Let $\mathbb{P} \in \mathcal{M}(\mu_1, \mu_2)$ be any admissible pricing measure. Since the super-replication constraint holds pointwise, it holds in expectation under $\mathbb{P}$:

$$\mathbb{E}^{\mathbb{P}}[\phi(S_1) + \psi(S_2) + \Delta(S_1)(S_2 - S_1)] \geq \mathbb{E}^{\mathbb{P}}[C(S_1, S_2)].$$

The martingale constraint $\mathbb{E}^{\mathbb{P}}[S_2 | S_1] = S_1$ implies, by the tower property,

$$\mathbb{E}^{\mathbb{P}}[\Delta(S_1)(S_2 - S_1)] = \mathbb{E}^{\mathbb{P}}[\Delta(S_1) \mathbb{E}^{\mathbb{P}}[S_2 - S_1 | S_1]] = 0.$$

This is the key role of the martingale constraint: it makes the dynamic trading term vanish in expectation, regardless of the choice of Δ . The expected cost of the strategy therefore reduces to

$$\mathbb{E}^{\mathbb{P}}[\phi(S_1)] + \mathbb{E}^{\mathbb{P}}[\psi(S_2)] = \int \phi d\mu_1 + \int \psi d\mu_2,$$

where we used the marginal constraints $S_t \sim \mu_t$.

2.2 Duality and MOT Formulation

Since the above holds for every $\mathbb{P} \in \mathcal{M}(\mu_1, \mu_2)$, we obtain the inequality

$$\inf_{\phi, \psi, \Delta} \left\{ \int \phi d\mu_1 + \int \psi d\mu_2 \right\} \geq \sup_{\mathbb{P} \in \mathcal{M}(\mu_1, \mu_2)} \mathbb{E}^{\mathbb{P}}[C(S_1, S_2)].$$

A fundamental result establishes that equality holds:

Theorem : Beiglböck–Henry-Labordère–Penkner 2013

Under mild regularity conditions on C and the marginals μ_1, μ_2 ,

$$\inf_{\phi, \psi, \Delta} \left\{ \int \phi d\mu_1 + \int \psi d\mu_2 \right\} = \sup_{\mathbb{P} \in \mathcal{M}(\mu_1, \mu_2)} \mathbb{E}^{\mathbb{P}}[C(S_1, S_2)].$$

The robust super-replication price therefore equals the worst-case expected payoff over all admissible pricing measures. Therefore, the robust pricing problem reduces to the **martingale optimal transport** (MOT) problem:

$$\sup_{\mathbb{P} \in \mathcal{M}(\mu_1, \mu_2)} \mathbb{E}^{\mathbb{P}}[C(S_1, S_2)]. \quad (1)$$

Explanation

Intuitively, the sup in (1) finds the joint coupling of S_1 and S_2 — consistent with market prices and no-arbitrage — that makes the exotic payoff as expensive as possible. The infimum on the left-hand side then finds the cheapest hedge that covers it.

The MOT problem (1) is generally intractable in closed form, even for simple payoff functions C . Solving it numerically requires discretizing the marginals and handling the martingale constraint efficiently. In what follows, we present in detail the theoretical foundations and numerical implementations of three approaches:

- **Linear programming formulation:** a direct discretization of (1) as a finite-dimensional linear program.
- **Semi-dual formulation:** a partial dualization of the primal problem that yields a convex minimization, solved numerically via neural network parametrization.
- **Entropic regularization:** a smoothed approximation that replaces the hard martingale constraint with a penalized objective, enabling fast Sinkhorn-type algorithms.

3 Linear approach

3.1 Definition of the Discrete Problem

To solve the problem numerically, we move from the continuous formulation to a discrete one by approximating the measures μ_1 and μ_2 with empirical measures $\hat{\mu}_1^n$ and $\hat{\mu}_2^n$.

3.1.1 Martingale Admissibility

According to Strassen's Theorem, the set $\mathcal{M}(\mu_1, \mu_2)$ is non-empty if and only if μ_1 and μ_2 are in convex order, denoted $\mu_1 \leq_{cx} \mu_2$. This condition is satisfied if:

1. Preservation of the mean: $\int x \mu_1(dx) = \int y \mu_2(dy) = S_0$
2. Convex inequality: $\int (x - k)^+ \mu_1(dx) \leq \int (y - k)^+ \mu_2(dy), \quad \forall k \in \mathbb{R}$

3.1.2 Construction of Empirical Measures (Hat Functions)

We use a discretization based on linear piecewise basis functions (hat functions). For a uniform grid $\{x_i\}_{i=1}^n$, the weights α_i^n (for μ_1) and β_j^n (for μ_2) are defined by:

$$\hat{\mu}_t^n\left(\frac{i}{n}\right) = n \int_{\left[\frac{i-1}{n}, \frac{i}{n}\right]} \left(x - \frac{i-1}{n}\right) \mu_t(dx) + n \int_{\left[\frac{i}{n}, \frac{i+1}{n}\right]} \left(\frac{i+1}{n} - x\right) \mu_t(dx) \quad (2)$$

Explanation

This method ensures better stability and preserves the moments of the distribution more accurately than a simple point-wise sampling.

3.2 Numerical Implementation and Relaxation

3.2.1 Dealing with unbounded support measures

To handle measures with unbounded support, we approximate μ by a truncated measure μ^R on $[0, R]$ defined via $G_R = \min(G, R)$ where $G \sim \mu$ and $G_R \sim \mu^R$.

Assuming a finite moment of order $\lambda > 1$, the approximation error in the 1-Wasserstein distance is bounded as follows:

$$W_1(\mu^R, \mu) \leq \mathbb{E}[(G - R)\mathbf{1}_{\{G > R\}}] \leq \frac{\mathbb{E}[G^\lambda]}{R^{\lambda-1}} \quad (3)$$

Defining $\delta(R) := \frac{\mathbb{E}[G^\lambda]}{R^{\lambda-1}}$, it follows that $W_1(\mu^R, \mu) \rightarrow 0$ as $R \rightarrow \infty$, ensuring the consistency of the truncated numerical problem.

3.2.2 Epsilon-Relaxation of the Martingale Constraint

In a discrete setting, the martingale equality $\mathbb{E}[S_2|S_1 = x_i] = x_i$ is often too restrictive and may lead to an empty feasible set due to discretization errors. We introduce a relaxation parameter ϵ such that:

$$\left| \sum_{j=1}^n p_{ij}(y_j - x_i) \right| \leq \epsilon \cdot \alpha_i, \quad \forall i \in \{1, \dots, m\} \quad (4)$$

The value of ϵ is optimized using a auxiliary parameter $\lambda > 1$. The algorithm implements a search for the "best" ϵ defined by the error function:

$$\epsilon(\lambda) = \frac{\mathbb{E}[|S_2|^\lambda]}{R^{\lambda-1}} + \frac{4}{n} \quad (5)$$

Explanation

The parameter ϵ balances two sources of error: truncation of the support and discretization of the measures. By tuning λ , we control this trade-off and select a relaxation level that is both theoretically justified and numerically effective.

3.2.3 Linear Programming Solver

The problem is solved using a Dual Simplex method (via the HiGHS solver). The algorithm follows these steps:

- **Mapping** $\epsilon(\lambda)$: Evaluating the error function over a grid of λ values.
- **Feasibility Test**: Iteratively searching for the optimal λ that minimizes ϵ while maintaining a feasible solution for the transport plan p_{ij} .
- **Optimization**: Maximizing the objective function $\sum_{i,j} p_{ij}c(x_i, y_j)$ subject to the relaxed martingale and marginal constraints.

4 The Semi-Dual Formulation of MOT

4.1 The MOT Problem (Recall)

Recall that we aim to find the maximum transport cost under the martingale constraint :

$$MK_c(\mu_1, \mu_2) := \sup_{\mathbb{P} \in \mathcal{M}} (\mu_1, \mu_2) \mathbb{E}^{\mathbb{P}}[c(S_1, S_2)]$$

A necessary condition for this problem to admit a solution was given by Strassen [1, Proof of Theorem 8]: the problem has a solution if and only if $\mu_1 \leq_{cx} \mu_2$ (convex order).

4.2 Kantorovich Dual for the MOT

Theorem [2, Proposition 4.4]: The dual formulation of the MOT problem is given by:

$$MK_c = \inf_{\substack{u \in L^1(\mu_2) \\ h \in C_b(\mathbb{R}^d, \mathbb{R}^d)}} \mathbb{E}^{\mu_2}[u(S_2)] + \mathbb{E}^{\mu_1}[\varphi(S_1)] \quad (6)$$

where φ is the c -martingale transform, defined as:

$$\varphi(s_1) := \sup_{s_2 \in \mathbb{R}} \left\{ c(s_1, s_2) - u(s_2) - h(s_1) \cdot (s_2 - s_1) \right\}$$

Proof Sketch: We start from the full Lagrangian of the primal MOT problem. The three constraints (marginals μ_1, μ_2 , and the martingale condition) yield three multipliers (u_1, u_2, h) . By strong duality (Fenchel-Rockafellar, see [3] for classic OT and Beiglböck et al. [2] for the martingale extension), the dual is written as:

$$MK_c = \inf_{u_1, u_2, h} \left\{ \mathbb{E}^{\mu_1}[u_1] + \mathbb{E}^{\mu_2}[u_2] \right\}$$

subject to the constraint $u_1(s_1) + u_2(s_2) + h(s_1)(s_2 - s_1) \geq c(s_1, s_2)$ for all (s_1, s_2) . For a fixed s_1 , the smallest value of $u_1(s_1)$ satisfying the constraint for all s_2 is:

$$u_1(s_1) \geq \sup_{s_2} \left\{ c(s_1, s_2) - u_2(s_2) - h(s_1)(s_2 - s_1) \right\}$$

The optimum is reached when the inequality is an equality. By setting $u := u_2$ and $\varphi(s_1) := \sup_{s_2} \{c - u - h(s_2 - s_1)\}$, we eliminate u_1 and obtain the semi-dual formulation (6). $\square$

4.3 Key Property: Convex Minimization

Proposition: The semi-dual problem is a convex minimization problem in (u, h) .

Proof: The functional to minimize is:

$$L(u, h) = \mathbb{E}^{\mu_2}[u(S_2)] + \mathbb{E}^{\mu_1} \left[\sup_{s_2} \{c(S_1, s_2) - u(s_2) - h(S_1)(s_2 - S_1)\} \right]$$

The first term is linear in u (thus convex). The second term is a supremum of affine functions in (u, h) : for each fixed s_2 , the expression $c(s_1, s_2) - u(s_2) - h(s_1)(s_2 - s_1)$ is affine in (u, h) . Since the supremum of convex functions is convex [4], and the expectation operator preserves convexity, the functional L is convex. $\square$

4.4 Gradient Computation via Danskin's Theorem

Theorem (Danskin [5], see also [6]): If $f(\theta, z) = \sup_{y \in Y} g(\theta, y)$ where g is differentiable with respect to θ and the supremum is attained at $y^*(\theta)$, then:

$$\nabla_{\theta} f = \nabla_{\theta} g(\theta, y^*(\theta))$$

This means we do not need to differentiate through the argmax; we simply evaluate the gradient at the point y^* that realizes the supremum.

Application: For our c -transform $\varphi(s_1)$, let us denote $s_2^*(s_1) = \operatorname{argmax}_{s_2} \{\dots\}$. Applying Danskin's theorem yields:

$$\frac{\partial L}{\partial \theta_u} = \mathbb{E}^{\mu_2} \left[\frac{\partial u}{\partial \theta}(S_2) \right] - \mathbb{E}^{\mu_1} \left[\frac{\partial u}{\partial \theta}(s_2^*(S_1)) \right]$$

(The minus sign appears because u has a negative coefficient inside the supremum).

$$\frac{\partial L}{\partial \theta_h} = -\mathbb{E}^{\mu_1} \left[(s_2^*(S_1) - S_1) \cdot \frac{\partial h}{\partial \theta}(S_1) \right]$$

The term $(s_2^* - S_1)$ is exactly the martingale residual. At the optimum, h adjusts the penalty so that $\mathbb{E}[s_2^* - S_1] \approx 0$.

4.5 Practical Computation of the c -transform

In practice, calculating the exact $\sup_{s_2}$ analytically is unfeasible. We approximate it by taking the maximum over a finite pool of N samples drawn from μ_2 :

$$\varphi(s_1) \approx \max_{j=1, \dots, N} \left\{ c(s_1, s_2^j) - u(s_2^j) - h(s_1)(s_2^j - s_1) \right\}$$

The larger N is, the better the approximation. In practice, a pool size of $K = 512$ is highly effective.

4.6 Numerical Resolution

4.6.1 Parametrization of Dual Variables

The dual potential u and the martingale multiplier h are parametrized by two independent neural networks u_{θ_u} and h_{θ_h} . This choice is grounded in the Universal Approximation Theorem [7, 8]: any continuous function on a compact subset of $\mathbb{R}$ can be approximated arbitrarily well by a feedforward network with at least one hidden layer and a non-polynomial activation, provided sufficient width.

4.6.2 Architecture

Both networks share the same architecture: two hidden layers of width $H = 32$, with a scalar input and scalar output. The activation function is Softplus,

$$\sigma(x) = \log(1 + e^x), \quad (7)$$

whose derivative is the Sigmoid function. This choice matters in practice: the c -transform requires evaluating the argmax over s_2 , and the smooth gradient flow of Softplus avoids the instabilities that would arise with ReLU at the kinks. Weights are initialized with the scheme $W \sim \mathcal{N}\left(0, \frac{2}{n_{\text{in}} + n_{\text{out}}}\right)$ to prevent early saturation.

4.6.3 Training Procedure

At each iteration, three independent batches are drawn: a minibatch $\{s_1^{(i)}\}_{i=1}^N \sim \mu_1$ and $\{s_2^{(i)}\}_{i=1}^N \sim \mu_2$ of size $N = 128$, plus a larger pool $\{\tilde{s}_2^{(j)}\}_{j=1}^K \sim \mu_2$ of size $K = 512$ used to discretize the supremum in the c -transform. The empirical loss is:

$$\hat{L}(\theta_u, \theta_h) = \frac{1}{N} \sum_{i=1}^N u_{\theta_u}(s_2^{(i)}) + \frac{1}{N} \sum_{i=1}^N \max_{1 \leq j \leq K} \left\{ c(s_1^{(i)}, \tilde{s}_2^{(j)}) - u_{\theta_u}(\tilde{s}_2^{(j)}) - h_{\theta_h}(s_1^{(i)}) (\tilde{s}_2^{(j)} - s_1^{(i)}) \right\}. \quad (8)$$

By Danskin's theorem, the gradients with respect to θ_u and θ_h are evaluated at the optimal $\tilde{s}_2^* = \operatorname{argmax}_j(\cdot)$, without differentiating through the max. Concretely, the gradient with respect to θ_u combines two backpropagation passes one through $u(s_2)$ with weight $+1$, and one through $u(\tilde{s}_2^*)$ with weight -1 while the gradient with respect to θ_h receives the residual $-(\tilde{s}_2^* - s_1)$ as upstream signal. Both networks are updated jointly using Adam [9] with learning rates $\eta_u = \eta_h = 5 \times 10^{-4}$ and gradient clipping at 5.0 to prevent instabilities. The training runs for 20000 iterations. At the end of training, we restore the best model from the last 1000 iterations, evaluated based on the minimum martingale error over a fixed set of 4096 points. We restricted our selection to this final window because the estimated optimal price remains stable over the last 5000 iterations. This stability allows us to safely choose the probability measure that minimizes the martingale error.

5 Entropic Relaxation of the Martingale OT Problem

We consider the entropy-regularized martingale optimal transport problem

$$\operatorname{MK}_\varepsilon(\mu^1, \mu^2) = \sup_{P \in \mathcal{M}(\mu^1, \mu^2)} \mathbb{E}^P[c(S_1, S_2)] - \varepsilon H(P|\rho^0), \quad (9)$$

where

$$H(P|\rho^0) = \mathbb{E}^P \left[\ln \left(\frac{dP}{d\rho^0} \right) - 1 \right]$$

is the relative entropy with respect to a prior measure ρ^0 . [10]

Explanation

The entropic regularization term is the entropy of the distribution p . Adding this term makes the problem strictly convex, guaranteeing the uniqueness of the solution and allowing the use of efficient iterative algorithms. The parameter $\varepsilon > 0$ controls the trade-off between minimizing the original cost and maximizing entropy. As $\varepsilon \rightarrow 0$ we recover the classical martingale OT problem.

5.1 Dual formulation

The problem can be rewritten using convex duality. [11]

Theorem 1 (Dual representation). *The entropic martingale OT problem admits the dual representation*

$$\begin{aligned} \operatorname{MK}_\varepsilon(\mu^1, \mu^2) = & \inf_{u_1, u_2, h} \mathbb{E}^{\mu^1}[u_1(S_1)] + \mathbb{E}^{\mu^2}[u_2(S_2)] \\ & + \varepsilon \mathbb{E}^{\rho^0} \left[\exp \left(\frac{c(S_1, S_2) - u_1(S_1) - u_2(S_2) - h(S_1)(S_2 - S_1)}{\varepsilon} \right) \right]. \end{aligned} \quad (10)$$

Proof : see Appendix 9. [12]

5.2 Optimality conditions

To compute the updates we differentiate the dual objective.
Define

$$J(u_1, u_2, h) = \mathbb{E}^{\mu^1}[u_1(S_1)] + \mathbb{E}^{\mu^2}[u_2(S_2)] + \varepsilon \mathbb{E} \rho^0 \left[e^{\frac{c(S_1, S_2) - u_1(S_1) - u_2(S_2) - h(S_1)(S_2 - S_1)}{\varepsilon}} \right]. \quad (11)$$

5.2.1 Gradient with respect to u_1

$$e^{-u_1(s_1)/\varepsilon} \int \rho^0(s_1, s_2) e^{\frac{c - u_2 - h(s_1)(s_2 - s_1)}{\varepsilon}} ds_2 = \mu^1(s_1).$$

Proof: see Appendix 9.

5.2.2 Gradient with respect to u_2

Similarly we obtain

$$e^{-u_2(s_2)/\varepsilon} \int \rho^0(s_1, s_2) e^{\frac{c(s_1, s_2) - u_1(s_1) - h(s_1)(s_2 - s_1)}{\varepsilon}} ds_1 = \mu^2(s_2). \quad (12)$$

5.2.3 Gradient with respect to h

The derivative with respect to $h(s_1)$ gives the martingale constraint

$$\int \rho^0(s_1, s_2)(s_2 - s_1) e^{\frac{c(s_1, s_2) - u_1(s_1) - u_2(s_2) - h(s_1)(s_2 - s_1)}{\varepsilon}} ds_2 = 0. \quad (13)$$

5.3 Numerical resolution

We now describe the numerical resolution of the dual problem using a Sinkhorn-type algorithm. The method is based on the first-order optimality conditions derived in the previous section.

Discrete setting

We discretize the problem by approximating the marginals μ^1 and μ^2 using Dirac masses supported on N points:

$$\mu^1 \approx \sum_{i=1}^N \mu_i^1 \delta_{s_1^i}, \quad \mu^2 \approx \sum_{j=1}^N \mu_j^2 \delta_{s_2^j}.$$

The reference density $\rho^0(s_1, s_2)$ is evaluated on the grid (s_1^i, s_2^j) .

The goal is to compute the dual potentials: $u_1(s_1)$, $u_2(s_2)$, $h(s_1)$, which enforce marginal constraints (via u_1, u_2) and martingale constraint (via h).

Key idea of the algorithm. These optimality equations are nonlinear and coupled. The Sinkhorn algorithm solves them by *alternating updates*, fixing some variables while solving for others [13]. In order to obtain a tractable numerical algorithm, these integrals are approximated by finite sums over the discretization grid.

Given grid points $(s_1^i)_{1 \leq i \leq N}$ and $(s_2^j)_{1 \leq j \leq N}$, we approximate

$$\int \rho^0(s_1, s_2) \varphi(s_1, s_2) ds_2 \approx \sum_{j=1}^N \rho^0(s_1, s_2^j) \varphi(s_1, s_2^j) w_j,$$

and similarly

$$\int \rho^0(s_1, s_2) \varphi(s_1, s_2) ds_1 \approx \sum_{i=1}^N \rho^0(s_1^i, s_2) \varphi(s_1^i, s_2) w_i,$$

where (w_i) and (w_j) are quadrature weights (typically uniform).

In practice, for simplicity, one often absorbs the weights into the definition of ρ^0 , leading to the approximation

$$\int \rho^0(s_1, s_2) \varphi(s_1, s_2) \approx \sum_{i,j} \rho_{i,j}^0 \varphi(s_1^i, s_2^j),$$

with $\rho_{i,j}^0 = \rho^0(s_1^i, s_2^j)$.

Algorithm.

1. Initialization.

Set $n = 0$ and initialize

$$u_1^{(0)} = 0, \quad u_2^{(0)} = 0, \quad h^{(0)} = 0.$$

2. Update of u_1 .

For each grid point s_1^i , compute $u_1^{(n)}(s_1^i)$ such that

$$\mu^1(s_1^i) = e^{-u_1^{(n)}(s_1^i)/\varepsilon} \sum_{j=1}^N \rho^0(s_1^i, s_2^j) \exp \left(\frac{c(s_1^i, s_2^j) - u_2^{(n-1)}(s_2^j) - h^{(n-1)}(s_1^i) \cdot (s_2^j - s_1^i)}{\varepsilon} \right).$$

3. Update of h .

For each s_1^i , compute $h^{(n)}(s_1^i)$ as the unique solution $\theta \in \mathbb{R}^d$ of

$$\sum_{j=1}^N \rho^0(s_1^i, s_2^j) (s_2^j - s_1^i) \exp \left(\frac{c(s_1^i, s_2^j) - u_2^{(n-1)}(s_2^j) - \theta \cdot (s_2^j - s_1^i)}{\varepsilon} \right) = 0.$$

4. Update of u_2 .

For each s_2^j , compute $u_2^{(n)}(s_2^j)$ such that

$$\mu^2(s_2^j) = e^{-u_2^{(n)}(s_2^j)/\varepsilon} \sum_{i=1}^N \rho^0(s_1^i, s_2^j) \exp \left(\frac{c(s_1^i, s_2^j) - u_1^{(n)}(s_1^i) - h^{(n)}(s_1^i) \cdot (s_2^j - s_1^i)}{\varepsilon} \right).$$

5. Iteration.

Set $n \leftarrow n + 1$ and repeat until convergence.

At convergence use the optimal $u_1(s_1)$, $u_2(s_2)$, $h(s_1)$, to compute the supremum

5.3.1 Reconstruction of the optimal plan

Once the dual variables are computed, the optimal coupling is recovered as

$$P_{ij} = P_{ij}^0 \exp \left(\frac{c(x_i, y_j) - u_1(x_i) - u_2(y_j) - h(x_i)(y_j - x_i)}{\varepsilon} \right).$$

This plan is then used to verify the marginal constraints and the martingale constraint.

5.3.2 Convergence criteria

The algorithm is stopped when the following residuals are sufficiently small:

$$\max_i \left| \sum_j P_{ij} - \alpha_i \right|, \quad \max_j \left| \sum_i P_{ij} - \beta_j \right|, \quad \max_i \left| \sum_j P_{ij}(y_j - x_i) \right|.$$

These quantities measure respectively the marginal errors for μ_1 and μ_2 , and the martingale error.

5.3.3 Numerical stability

Because the algorithm involves repeated exponentials, direct evaluation may suffer from overflow or underflow when ε is small. To avoid this issue, we compute all sums using the log-sum-exp trick. This stabilizes the Sinkhorn iterations and improves robustness for small regularization parameters.

5.3.4 Influence of the regularization parameter

The parameter ε controls the trade-off between accuracy and smoothness. Large values of ε lead to faster convergence but a more diffuse coupling, while smaller values produce a solution closer to the original MOT problem at the cost of greater numerical instability.

6 Numerical Results

This section presents a numerical comparison between the three approaches used to solve the martingale optimal transport problem: the linear programming formulation, the semi-dual formulation, and the entropic penalty. The comparison is carried out in terms of the resulting option prices and the structure of the optimal transport plans. We also study the sensitivity of the methods with respect to the choice of payoff functions.

This numerical results presented are with regards to marginals defined using two uniform measures :

$$\begin{aligned} S_1 &\sim \mathcal{U}(1, 3) \\ S_2 &\sim \mathcal{U}(0, 4) \end{aligned}$$

For each payoff, we compute:

- the robust price obtained by linear programming,
- the price obtained from the semi-dual formulation,
- the price obtained from the entropic penalty,
- the associated transport plan matrices.

6.1 result comparisons

Absolute difference between prices: We will first consider the results for the cost function defined as followed

$$c(S_1, S_2) = |S_2 - S_1|$$

The results of the optimal price that we get are shown in this table, the relaxation parameter ϵ for the entropic penalty was taken to be 0.005, same for all future tests.

LP Price	Semi-Dual Price	Entropic Price
1.13	0.99	0.99

Table 1: Comparison of robust prices obtained with the LP, semi-dual, and entropic methods.

The transport plans obtained by the three methods are as followed:

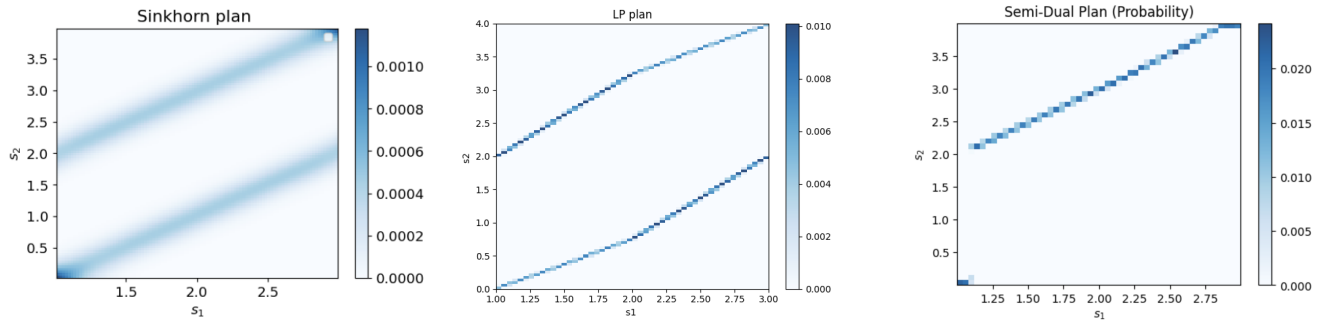


Figure 1: Optimal transport matrices obtained.

Asian Option: This time we will consider the same example and parameters, except we will use the cost function

$$c(S_1, S_2) = \left(\frac{S_1 + S_2}{2} - K \right)^+$$

with a strike of $K = 1.5$.

The results of the optimal price that we get are shown in this table:

LP Price	Semi-Dual Price	Entropic Price
0.66	0.66	0.65

Table 2: Comparison of robust prices obtained with the LP, semi-dual, and entropic methods.

The transport plans obtained by the three methods are as followed:

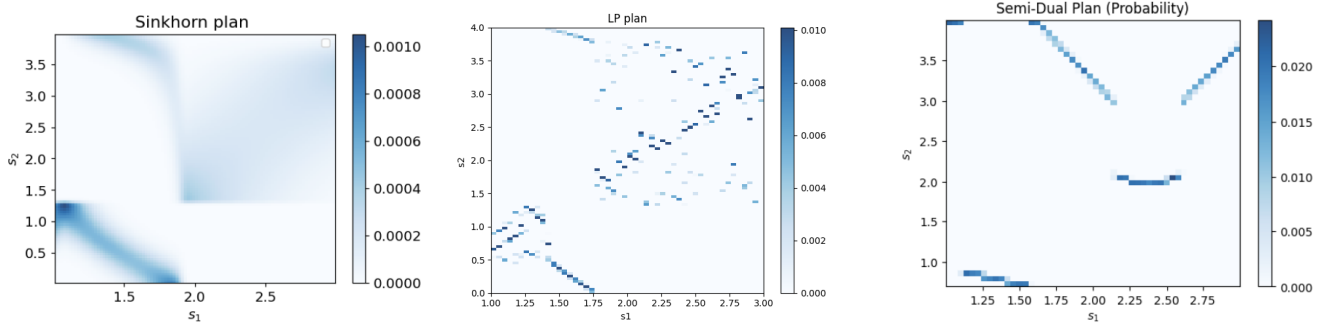


Figure 2: Optimal transport matrices obtained for an Asian option.

Lookback Option: Finally, let's consider a lookback option characterized by the following cost function:

$$c(S_1, S_2) = (\max(S_2, S_1) - K)^+$$

The results of the optimal price that we get are shown in this table:

LP Price	Semi-Dual Price	Entropic Price
1.09	1.04	1.02

Table 3: Comparison of robust prices obtained with the LP, semi-dual, and entropic methods.

the transport plan obtained by the three methods are as followed

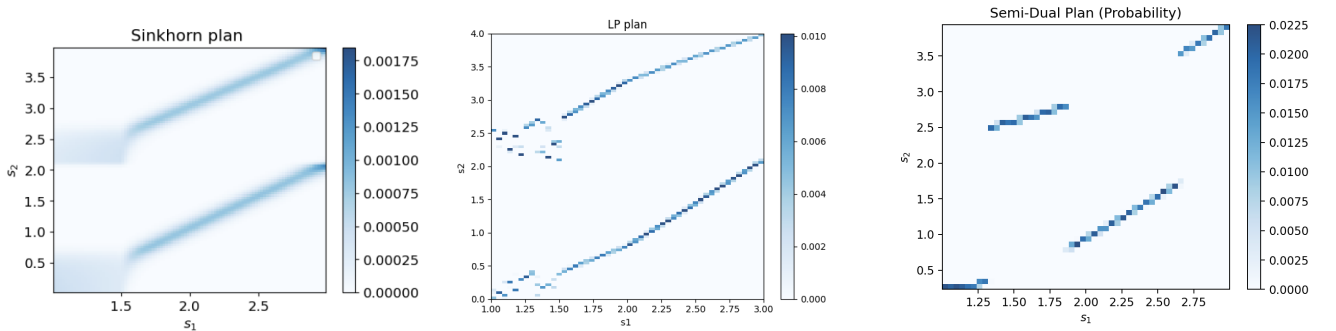


Figure 3: Optimal transport matrices obtained for a lookback option.

7 Discussion

7.1 Strengths and Weaknesses of the Three Approaches

Linear programming yields the exact solution (up to discretization) and serves as a benchmark. However, as pointed out by Henry-Labordère [13, Section 2.1], the complexity is $\mathcal{O}(N^3)$ for the Hungarian algorithm, which restricts this approach to modest grid sizes ($m \times n \leq 100 \times 100$). For problems in dimension $d \geq 2$, the number of constraints explodes, rendering the method inapplicable.

Entropic relaxation converges rapidly thanks to Sinkhorn's algorithm but introduces a systematic bias of $\mathcal{O}(\varepsilon)$ [11, 14]. Numerical results confirm this trade-off: with $\varepsilon = 0.005$, the

obtained prices are close to the LP results but not identical. Furthermore, as reported by Henry-Labordère [13, Section 2.3] and Guo-Obłój [15], computations become numerically unstable when $\varepsilon \rightarrow 0$, due to the exponential term $e^{c/\varepsilon}$ blowing up. The log-sum-exp trick mitigates this issue but does not entirely eliminate it.

The semi-dual formulation is the most efficient in practice: it requires only two neural networks, relies on convex minimization, and achieves fast convergence thanks to Danskin's theorem [5]. Numerical results show prices consistent with the LP approach for the three tested payoffs (Asian call, Lookback, absolute value).

7.2 Structural Limitation of the Semi-Dual Formulation

The semi-dual formulation, as implemented, yields a unique $s_2^*(s_1) = \operatorname{argmax}_{s_2} \{c(s_1, s_2) - u(s_2) - h(s_1)(s_2 - s_1)\}$ for each s_1 . This means the reconstructed transport plan takes the form:

$$P^*(s_1, s_2) = \mu_1(ds_1) \cdot \delta(s_2 - s_2^*(s_1))$$

This is a deterministic transport: each s_1 is mapped to a single point s_2^* . However, the Beiglböck-Juillet theorem [16] demonstrates that under the martingale twist condition, the MOT solution is supported on two maps:

$$P^*(s_1, s_2) = q(s_1)\delta(s_2 - T_u(s_1)) + (1 - q(s_1))\delta(s_2 - T_d(s_1))$$

with $T_d(s_1) \leq s_1 \leq T_u(s_1)$. A deterministic transport cannot satisfy the martingale constraint $\mathbb{E}[S_2|S_1] = S_1$ in a non-trivial way: if $s_2^*(s_1)$ is a single point, then either $s_2^* = s_1$ (trivial, sub-optimal cost), or the martingale condition is violated.

In practice, the discrete c-transform (taking the maximum over a finite pool of K points) partially bypasses this issue: the argmax can jump between different s_2^j depending on the batch, producing an implicit randomization that mimics the two branches. This is observed in the semi-dual transport plans (Figures 1-3): the points do not form a single sharp curve but scatter around two branches, albeit less cleanly than in the LP approach.

This limitation explains the slight price discrepancies observed between the LP and the semi-dual methods (e.g., 1.13 vs. 0.99 for $|S_2 - S_1|$, or 1.09 vs. 1.04 for the Lookback option). The semi-dual approach underestimates the price because it fails to perfectly capture the two-branch structure.

7.3 Avenues for Improvement

A natural extension would be to modify the c-transform to explicitly return the two best s_2 candidates instead of a single argmax, and to optimize the weight q as an additional variable. This would align with the structure of Henry-Labordère's 5-network primal-dual algorithm [13, Section 3.8], but within a minimization framework (no min-max), which is potentially more stable.

Another direction would be to use the semi-dual formulation as an initialization for the 5-network primal-dual algorithm: the potentials u and h learned by the semi-dual method would provide a strong starting point, significantly reducing the number of iterations required for the min-max convergence.

8 Conclusion

This work implemented and compared three numerical approaches for martingale optimal transport applied to the robust pricing of exotic options. Linear programming provides the reference solution but does not scale. Entropic relaxation is fast but biased. The semi-dual formulation, based on Kantorovich duality [2] and Danskin's theorem [5], offers the best compromise between efficiency and accuracy, requiring only two neural networks and ensuring fast convergence via convex minimization.

The three methods yield consistent prices across the tested payoffs (Asian call: 0.65-0.66, Lookback: 1.02-1.09, Absolute value: 0.99-1.13), validating the robust model-free approach for pricing exotic options.

However, the semi-dual formulation reconstructs a deterministic transport (a single argmax), whereas the optimal MOT structure under the martingale twist condition [16] requires two transport maps. This structural limitation, identified in our analysis, paves the way for future extensions combining the stability of the semi-dual approach with the structural richness of the primal-dual algorithm [13].

9 Appendix

9.1 Fenchel-Rockefeller dual problem computation

Fenchel–Rockafellar duality

We recall the classical **Fenchel–Rockafellar duality theorem** in a general setting. Let X, Y be Banach spaces, $A : X \rightarrow Y$ a continuous linear operator, and let $F : X \rightarrow \mathbb{R} \cup \{+\infty\}$ and $G : Y \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper convex lower semicontinuous functions. Consider the primal problem

$$\inf_{x \in X} F(x) + G(Ax).$$

The associated dual problem is

$$\sup_{y^* \in Y^*} -F^*(A^*y^*) - G^*(-y^*),$$

where:

- F^* and G^* denote the convex conjugates,
- $A^* : Y^* \rightarrow X^*$ is the adjoint operator.

Convex reformulation. We start from the entropy-regularized martingale optimal transport problem

$$\sup_{P \in \mathcal{M}(\mu^1, \mu^2)} \left(\mathbb{E}^P[c(S_1, S_2)] - \varepsilon H(P \mid \rho^0) \right),$$

where the relative entropy with respect to a reference measure ρ^0 is defined by

$$H(P \mid \rho^0) = \int \left(\ln \frac{dP}{d\rho^0} - 1 \right) dP.$$

In order to apply Fenchel–Rockafellar convex duality, it is convenient to perform a change of variable and rewrite the optimization problem over densities with respect to the reference

measure ρ^0 . Assuming that P is absolutely continuous with respect to ρ^0 , we introduce the Radon–Nikodym density

$$f = \frac{dP}{d\rho^0}.$$

Under this change of variable, the measure P can be written as

$$dP = f d\rho^0.$$

Substituting this representation into the objective functional yields

$$\mathbb{E}^P[c(S_1, S_2)] = \int c(s_1, s_2) f(s_1, s_2) d\rho^0,$$

and

$$H(P \mid \rho^0) = \int (f \ln f - f) d\rho^0.$$

Therefore the objective becomes

$$\int \left(c(s_1, s_2) f(s_1, s_2) - \varepsilon \left(f(s_1, s_2) \ln f(s_1, s_2) - f(s_1, s_2) \right) \right) d\rho^0.$$

For convenience we introduce the convex function

$$F(x) = \varepsilon(x \ln x - x),$$

so that the objective functional can be written compactly as

$$\int \left(c(s_1, s_2) f(s_1, s_2) - F(f(s_1, s_2)) \right) d\rho^0.$$

Linear formulation of the constraints. The constraints defining the set $\mathcal{M}(\mu^1, \mu^2)$ can now be expressed as linear constraints on the density f .

First, the marginal constraints become

$$\int f(s_1, s_2) \rho^0(ds_2 \mid s_1) = \mu^1(s_1),$$

and

$$\int f(s_1, s_2) \rho^0(ds_1 \mid s_2) = \mu^2(s_2).$$

Second, the martingale constraint

$$\mathbb{E}^P[S_2 - S_1 \mid S_1] = 0$$

is equivalent to

$$\int (s_2 - s_1) f(s_1, s_2) \rho^0(ds_2 \mid s_1) = 0.$$

These constraints can be conveniently summarized by introducing a linear operator

$$\Delta(f) = \left(\int f(s_1, s_2) \rho^0(ds_2 \mid s_1), \int f(s_1, s_2) \rho^0(ds_1 \mid s_2), \int (s_2 - s_1) f(s_1, s_2) \rho^0(ds_2 \mid s_1) \right).$$

The feasibility conditions then become

$$\Delta(f) = (\mu^1, \mu^2, 0).$$

Convex optimization formulation. Using this notation, the primal problem can be rewritten as a convex optimization problem of the form

$$\inf_f F^\rho(f) + G(\Delta f),$$

where

$$F^\rho(f) = \int (F(f) - cf) d\rho^0$$

and

$$G(y) = \begin{cases} 0 & \text{if } y = (\mu^1, \mu^2, 0), \\ +\infty & \text{otherwise.} \end{cases}$$

This formulation fits exactly into the Fenchel–Rockafellar convex duality framework. In particular, the operator Δ is linear and the functional F^ρ is convex by construction (since $F(x) = \varepsilon(x \ln x - x)$ is convex on $\mathbb{R}_+$). Consequently, the dual problem can be obtained by applying Fenchel–Rockafellar duality to the pair of convex functions F^ρ and G composed with the linear operator Δ .

Derivation of the dual problem. We now apply Fenchel–Rockafellar duality to the problem

$$\inf_f F^\rho(f) + G(\Delta f).$$

The dual problem is given by

$$\sup_\lambda -(F^\rho)^*(\Delta^* \lambda) - G^*(-\lambda),$$

where $\lambda = (u_1, u_2, h)$ belongs to the dual space.

Step 1: Adjoint operator.

Recall that

$$\Delta(f) = \left(\Delta_1(f), \Delta_2(f), \Delta_3(f) \right),$$

with

$$\begin{aligned} \Delta_1(f)(s_1) &= \int f(s_1, s_2) \rho^0(ds_2 \mid s_1), \\ \Delta_2(f)(s_2) &= \int f(s_1, s_2) \rho^0(ds_1 \mid s_2), \\ \Delta_3(f)(s_1) &= \int (s_2 - s_1) f(s_1, s_2) \rho^0(ds_2 \mid s_1). \end{aligned}$$

The adjoint operator Δ^* acts as

$$\Delta^*(u_1, u_2, h)(s_1, s_2) = u_1(s_1) + u_2(s_2) + h(s_1)(s_2 - s_1).$$

Step 2: Conjugate of G .

By definition of G , we obtain

$$G^*(-\lambda) = -\langle \lambda, (\mu^1, \mu^2, 0) \rangle = -\mathbb{E}^{\mu^1}[u_1(S_1)] - \mathbb{E}^{\mu^2}[u_2(S_2)].$$

Step 3: Conjugate of F^ρ .

We compute

$$(F^\rho)^*(\phi) = \int F^*(\phi(s_1, s_2) + c(s_1, s_2)) d\rho^0,$$

where $F(x) = \varepsilon(x \ln x - x)$ has conjugate

$$F^*(y) = \varepsilon e^{y/\varepsilon}.$$

Therefore,

$$(F^\rho)^*(\Delta^* \lambda) = \varepsilon \int \exp \left(\frac{c(s_1, s_2) - u_1(s_1) - u_2(s_2) - h(s_1)(s_2 - s_1)}{\varepsilon} \right) d\rho^0.$$

Step 4: Dual formulation.

Putting everything together, we obtain the dual problem

$$\begin{aligned} & \inf_{u_1, u_2, h} \mathbb{E}^{\mu^1}[u_1(S_1)] + \mathbb{E}^{\mu^2}[u_2(S_2)] \\ & + \varepsilon \mathbb{E}^{\rho^0} \left[\exp \left(\frac{c(S_1, S_2) - u_1(S_1) - u_2(S_2) - h(S_1)(S_2 - S_1)}{\varepsilon} \right) \right], \end{aligned}$$

which coincides with the stated dual formulation.

9.2 Computation of gradient

Let φ be a smooth test function. We compute the Gâteaux derivative

$$\left. \frac{d}{dt} J(u_1 + t\varphi, u_2, h) \right|_{t=0}.$$

Only two terms depend on u_1 .

First term:

$$\frac{d}{dt} \int (u_1 + t\varphi) \mu^1 = \int \varphi(s_1) \mu^1(s_1) ds_1.$$

Second term:

Inside the exponential we have

$$\Phi_t = \Phi - t\varphi(s_1),$$

so

$$\frac{d}{dt} e^{\frac{\Phi_t}{\varepsilon}} = -\frac{1}{\varepsilon} \varphi(s_1) e^{\frac{\Phi}{\varepsilon}}.$$

Therefore,

$$\begin{aligned} \frac{d}{dt} \varepsilon \int \rho^0 e^{\frac{\Phi_t}{\varepsilon}} &= \varepsilon \int \rho^0 \left(-\frac{1}{\varepsilon} \varphi(s_1) \right) e^{\frac{\Phi}{\varepsilon}} \\ &= - \int \varphi(s_1) \left(\int \rho^0(s_1, s_2) e^{\frac{\Phi(s_1, s_2)}{\varepsilon}} ds_2 \right) ds_1. \end{aligned}$$

Total variation:

$$\delta J = \int \varphi(s_1) \left[\mu^1(s_1) - \int \rho^0(s_1, s_2) e^{\frac{\Phi}{\varepsilon}} ds_2 \right] ds_1.$$

Hence by definition,

$$\frac{\delta J}{\delta u_1}(s_1) = \mu^1(s_1) - \int \rho^0(s_1, s_2) e^{\frac{\Phi(s_1, s_2)}{\varepsilon}} ds_2.$$

Factoring $e^{-u_1(s_1)/\varepsilon}$ yields

$$e^{-u_1(s_1)/\varepsilon} \int \rho^0(s_1, s_2) e^{\frac{c - u_2 - h(s_1)(s_2 - s_1)}{\varepsilon}} ds_2 = \mu^1(s_1).$$

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